CAREFLOW

WEEK 4
PROJECT PROGRESS REPORT

Clinical Pathway Process Mining

Project domain	Healthcare Data Analytics and Process Mining
Reporting period	Week 4
Implementation focus	Clinical pathway conformance checking
Technology used	Python, Pandas, CSV and XES event logs

This report documents the Week 4 implementation work, conformance analysis results, automated testing, deliverables, and remaining activities for the CareFlow project.

1. Introduction

During Week 4, the project focused on implementing conformance checking for CareFlow. The objective was to compare actual patient journeys with an expected clinical pathway, identify deviations from the defined process rules, and evaluate how consistently patient cases followed the reference sequence of healthcare activities. This phase extended the earlier process discovery work by assessing actual activity sequences against an explicitly defined analytical reference model.

2. Defining the Expected Clinical Pathway

The following reference pathway was defined for the conformance analysis:

Registration → Triage → Doctor Consultation → Optional Laboratory Test / X-Ray → Doctor Consultation → Treatment → Discharge

- Laboratory tests and X-rays were treated as optional because diagnostic requirements may vary between patients.
- Repeated triage and doctor consultation activities were permitted to account for possible reassessments.

This pathway is an analytical reference model created for this project; it is not intended to represent a universal clinical guideline.

3. Implementation of the Conformance Checking Module

A Python-based conformance checking module was developed to compare the activity sequence in each patient case with the defined pathway rules. The checker evaluates:

- Whether each patient journey begins with Registration.
- Whether each patient journey ends with Discharge.
- Whether consecutive activities follow permitted transition rules.
- Whether diagnostic activities are followed by a doctor consultation before treatment or discharge.
- Whether repeated activities comply with the rules defined for the analysis.

The module automatically identifies unexpected transitions and records cases in which one or more reference rules are not satisfied.

4. Dataset Analyzed

The checker was executed against the cleaned CareFlow event log.

Dataset metric	Total
Patient cases	500
Event records	3,472
Distinct activity types	7

5. Conformance Checking Results

All 500 patient cases were evaluated using the implemented Python rules. The key results are summarized below.

500	220	280	44.00%
PATIENT CASES	CONFORMING CASES	NON-CONFORMING	CASE CONFORMANCE

Evaluation metric	Result
Total patient cases analyzed	500
Conforming cases	220
Non-conforming cases	280
Case-level conformance rate	44.00%
Observed transitions	2,972
Allowed transitions	2,603
Unexpected transitions	369
Transition-level conformance rate	87.58%

The results show that 220 patient cases satisfied all defined conformance rules, while 280 cases contained at least one deviation. At the transition level, 87.58% of observed transitions followed the permitted rules. The remaining transitions were flagged for further investigation. A flagged deviation is an analytical finding and does not, by itself, establish a clinical error or poor-quality care.

6. Identification of Process Deviations

The module generated detailed outputs to support investigation of differences between actual journeys and the reference pathway:

- Case-level conformance results: indicates whether each patient case conforms to the defined rules.
- Transition-level conformance results: records observed activity transitions and their conformance status.
- Conformance violations: lists unexpected transitions and the associated patient cases.
- Conformance summary: provides overall results and calculated metrics.
- Detailed conformance report: summarizes the analysis and its findings.

7. Automated Testing and Validation

Six out of six automated unit tests passed successfully. The tests validated the conformance checking logic and its handling of the defined process rules. The generated script was also executed independently and produced consistent results.

8. Project Deliverables Completed

- Python-based conformance checking script.
- Case-level conformance results in CSV format.
- Transition-level conformance results in CSV format.
- Conformance violation report.
- Conformance summary in JSON format.
- Detailed conformance analysis report.
- Automated unit test results.
- Complete implementation package in ZIP format.

9. Challenges and Observations

A key implementation challenge was defining clear rules for optional diagnostic activities and repeated clinical activities. Since patients may follow different journeys, the reference pathway needed to allow reasonable variations

while still identifying unexpected transitions. A process deviation should be treated as a finding for further review, rather than an automatic indication of clinical inefficiency.

10. Pending Activities and Future Improvements

- Integration and validation of the analysis using PM4Py.
- Execution and validation of the BigQuery and dbt pipeline.
- Integration of conformance results into the final dashboard.
- Dashboard filtering and drill-down validation.
- Preparation of final screenshots and project documentation.
- Final project demonstration and technical viva preparation.

The current implementation is a Python rule-based conformance checker. It is not PM4Py token replay or alignment-based conformance checking.

11. Conclusion

Week 4 was an important implementation phase of the CareFlow project. We defined an expected clinical pathway, developed a Python-based conformance checking module, and evaluated all 500 patient cases containing 3,472 event records. The analysis identified 220 conforming cases and 280 non-conforming cases, producing a case-level conformance rate of 44.00%. At the transition level, the conformance rate was 87.58%. Six automated unit tests were completed successfully, and the supporting reports and result files were generated.

Overall, Week 4 established a working foundation for evaluating patient journey deviations and identifying areas requiring further investigation. The completed implementation will support the remaining integration, dashboard validation, and final project presentation activities.

END OF WEEK 4 PROJECT PROGRESS REPORT